

OrcaTrade

The import-operations team, available 24/7.

A complete account of what we have built: the platform, how the AI agents operate, how and where we hold data, the security and compliance posture, and the road to a category-defining trade-compliance business.

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Scope: production system as deployed. Every figure in this document is drawn from the live codebase and test suite.

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One-sentence thesis. OrcaTrade pairs deterministic, audited trade-compliance calculators with a suite of AI agents that reason over them — so European importers get the judgement of a senior customs/logistics/finance team, grounded in numbers a regulator or auditor can verify, at software margins.

The business at a glance

OrcaTrade is an AI-native trade-compliance and import-operations platform for European SMEs sourcing from Asia. Five domains — search, sourcing, intelligence (compliance), logistics, finance — delivered as one platform that produces recommendations the user can *trust*, because every number is computed by audited code, never invented by a language model.

5+1

AI AGENTS + ORCHESTRATOR

2,949

AUTOMATED TESTS, ALL GREEN

727

INDEXED WEB PAGES

10

DETERMINISTIC CALCULATORS

39,813

LIVE SANCTIONS DESIGNATIONS

3

LANGUAGES (EN / PL / DE)

What we have built (production today)

- **A trustworthy decision engine.** Ten deterministic calculators cover EU/UK customs & duty, anti-dumping/countervailing duties, preferential origin, CBAM, EUDR, REACH, CE marking, FX risk, freight routing, warehousing, working capital and total cost of ownership.
- **Six AI agents** (compliance, logistics, sourcing, finance, a meta-orchestrator, and a personal agent that reasons over the signed-in user's own data) that call the calculators as tools and write the prose on top, with citations.
- **A proactive monitoring agent** that watches every saved plan and flags cost drift, FX exposure, compliance deadlines and sanctions changes — automatically.
- **A durable data spine** on Postgres + Redis, with GDPR-grade privacy, a tamper-evident audit trail, and live denied-party screening against the OFAC, UK OFSI and UN consolidated lists.
- **An acquisition moat** of 727 indexed pages including 658 localised SEO guides across three languages.

Why it matters

Import compliance is high-stakes and getting harder: CBAM, EUDR, anti-dumping stacks and sanctions regimes change constantly, and a single mistake can mean a seized container, a back-dated duty bill, or a sanctions breach. SMEs cannot afford a full in-house trade team. OrcaTrade is that team — as software — and the combination of (a) a growing corpus of regulatory data, (b) AI agents tuned and evaluated against it, and (c) a connected base of importer accounts compounds into a defensible moat over time.

Build posture. The company is deliberately investing now to lift earlier pre-funding constraints — promoting Postgres to the durable corpus, adopting frontier models for customer-facing reasoning, and building the enterprise-trust surfaces that remove buyer objections. This brief documents the system as it stands and the plan from here.

What OrcaTrade is — five domains, one platform

An importer comes to OrcaTrade with a question — "what will it really cost to bring 1,000 e-bikes from China into Poland, and what am I legally on the hook for?" The platform answers it end-to-end, across five connected domains.

The five domains

DOMAIN	WHAT IT ANSWERS	BACKED BY
Search	Find the right HS commodity code, supplier archetype, and starting assumptions.	Curated HS lookup, guides corpus
Sourcing	Supplier evaluation, MOQ & FOB economics, sample logistics, factory scoring.	sourcing-quote , sample-quote
Intelligence (compliance)	Duty, anti-dumping/CVD, preferential origin, CBAM, EUDR, REACH, CE — what applies and what it costs.	customs-quote + regime DBs
Logistics	Sea/air mode choice, lane routing, bonded warehousing, customs clearance.	routing-quote , warehouse-quote
Finance	Payment instruments, FX risk & hedging, working-capital cycle, insurance, returns, total cost of ownership.	finance-quote , fx-quote , tco-quote

The flagship surface — the Import Plan Builder

The interactive `/start/` wizard (also `/pl/start/` , `/de/start/`) takes a handful of inputs — product, origin, destination, value, weight, payment terms — and runs the full calculator chain to produce a single, shareable **import plan**: landed cost broken down to the cent, duty and AD/CVD stacking, preferential-origin alternatives, compliance obligations, freight options, working-capital impact and an annual total-cost-of-ownership view. Plans can be saved, compared across origins, re-run under alternate scenarios, exported to PDF, and shared by link.

The conversational surface — the agents

On top of the structured wizard, the user can simply *ask*. The orchestrator agent routes the question across all five domains, calls whichever calculators are relevant, and returns a grounded answer with citations and a clear recommendation — escalating to a human before anything irreversible.

Coverage today. EU & UK customs and duty · anti-dumping / countervailing duties · preferential origin (EBA, GSP/GSP+, EVFTA, EUKFTA, EUJEPa, A.TR) · CBAM · EUDR · REACH · CE marking (LVD/EMC/RED, machinery) · RoHS · WEEE · batteries · toys · cosmetics · GPSR · PPWR · footwear labelling · FX across 13 currencies · sea/air freight · bonded warehousing · working capital · TCO.

Calculator-grounded AI — why our numbers can be trusted

The single most important design decision in the company: **the language model never produces a number that drives a business decision**. Every figure — duty, landed cost, FX risk, revenue, deadline — is computed by deterministic code. The AI writes the explanation on top, and must cite the calculator that produced each number.

The contrast that wins deals. A generic AI chatbot will confidently hallucinate a duty rate. OrcaTrade physically cannot: the duty rate comes from an audited calculator, the AI is only allowed to narrate it, and an automated test parses every number in the AI's output and fails the build if any of them doesn't trace to a calculator result.

How it is enforced (not just promised)

1. All money/quote maths live in dedicated calculators (`lib/intelligence/*-quote.js`). They contain no AI code.

2. The AI SDK may only be imported in the agent/handler layer — never in a calculator. Calculators stay provably LLM-free.

3. Integer-cents money. No floating-point arithmetic on money anywhere in the calculators; a shared money module uses banker's rounding and converts only at the boundaries.

4. Citations are required. The agents cite a `[chunk-id]` for every regulatory claim and the tool for every number, with explicit confidence tiers.

5. Grounding is tested. The eval scorer extracts every monetary/percentage token from an agent's answer and asserts each matches a calculator output within tolerance — an ungrounded number fails the case.

What the AI is for

Judgement and language: understanding a messy real-world question, choosing which calculators to run and in what order, weighing trade-offs ("hedge or accept this FX exposure?"), explaining a result in plain UK English across three languages, and knowing when to stop and escalate to a human. That is genuinely valuable — and it is safe precisely because the numbers underneath it are not the model's to invent.

Why this is the moat, not a constraint. It lets us deploy frontier models aggressively on the *reasoning* while keeping the *arithmetic* auditable. Regulators, insurers and enterprise buyers can be shown exactly where every number came from — which is the entire basis of trust in this market.

How the platform is built

OrcaTrade runs on **Vercel** with auto-deploy on every push to `main`. The back end is a lean, dependency-light Node.js codebase; the front end is a fast static site plus the interactive wizard, with a full application shell on the near-term roadmap.

The request path

```

browser / API client
|
▼
api/[...path].js - single Vercel function, ~50 logical endpoints
|   routes /api/customs → customs handler, /api/orchestrator → meta-agent, ...
▼
lib/handlers/* - request handling, auth, the AI agents
|
├─ lib/intelligence/*-quote.js - deterministic calculators (no AI)
├─ lib/ai/* - the only place the AI SDK is imported
├─ KV (Upstash Redis) - sessions, cache, rate limits, hot state
└─ Neon Postgres - durable corpus (dual-write)

```

Key architectural choices

Consolidated router

All ~50 endpoints dispatch through one serverless function. URL semantics are preserved (`/api/customs` still means customs). This kept the system within platform function limits while staying simple to reason about.

Versioned API contracts

Public responses are served under a frozen `/api/v1/` contract; breaking changes go to a new version, so integrations never break silently.

Lean dependency surface

Only two runtime dependencies — the Anthropic AI SDK and the Neon Postgres driver. A small attack surface, fast cold starts, and no build-pipeline fragility.

Resilient by default

Every external HTTP call is wrapped with a timeout, a fallback and retries via a circuit breaker, and logged through a structured logger. An upstream outage degrades gracefully rather than failing the user.

Scheduled work

Background jobs run on a token-gated cron dispatcher driven by GitHub Actions — twelve jobs today, including the nightly tariff-cache warm, the weekly sanctions-list refresh, the regulatory-drift watcher, the compliance-deadline reminders, and the new proactive monitoring scan. Every run records its outcome to a dashboard so "did Monday's job actually fire?" is always answerable.

Graceful degradation is a theme. No database configured? The system serves sample data and never crashes. No embeddings key? Retrieval falls back to keyword search. A sanctions refresh missed? Screening falls back to a bundled list and says so. The platform is built to keep working when a piece is missing.

How & where we hold data

OrcaTrade runs a two-tier data spine: a fast key-value store for hot, ephemeral state, and a durable Postgres corpus for everything that must survive and be queried across users. Critical records are written to *both* (dual-write), with the key-value store as the synchronous primary and Postgres as the durable backstop.

Tier 1 — KV (Upstash Redis): fast & ephemeral

Holds session cookies, magic-link tokens, rate-limit counters, the warmed tariff cache, circuit-breaker state, and the hot copy of user records (saved plans, portfolios, alerts, agent memory). Reads and writes are sub-millisecond; entries carry sensible TTLs.

Tier 2 — Neon Postgres: durable corpus

The system of record that outlives KV's TTLs and supports cross-user analytics. Schema is migration-tracked (auto-discovered, versioned, idempotent). Current tables:

TABLE	HOLDS	PRIVACY
users / organisations / memberships	Accounts, orgs, seats	email_hash only
saved_plans / saved_portfolios	Import plans & multi-SKU catalogues	email_hash only
actuals	Real outcomes vs forecasts (calibration)	email_hash only
monitoring_alerts	Proactive monitoring alerts	email_hash only
agent_memory	Per-user facts the agent remembers	email_hash only
corpus_chunks	Regulatory text + vector embeddings (RAG)	no PII
sanctions_entries / sanctions_refresh	OFAC/OFSI/UN designations + freshness	public data
audit_log / events	State-change & analytics trail	pseudonymised
subscriptions / prompt_runs	Billing state · AI cost telemetry	email_hash only

The dual-write pattern

```
save a plan
|
├─ KV      write the full record (synchronous – the UI reads this)
└─ Postgres mirror it, best-effort (durable corpus; a DB blip never breaks the save)
```

The KV write is authoritative and returned immediately; the Postgres mirror is fire-and-forget and can never break a flow that already succeeded. This is how we get both speed and durability without a heavy ORM or transaction coupling.

Direction of travel. The roadmap promotes Postgres to the primary store as volumes grow, with the durable corpus already in place via dual-write today — so the migration is an evolution, not a rebuild.

How we protect personal data

Privacy is built into the data model, not bolted on. The guiding rule: **no raw personal data in Postgres or in the event log**. A user's email address is stored only as a one-way hash; the raw address lives only where it is operationally required (the session cookie and the magic-link/email service).

How identity is handled

- **email_hash everywhere durable**. Postgres and the event log key users by a SHA-256-derived hash, never the raw address. Analytics work; re-identification does not.
- **Passwordless auth**. Sign-in is by magic link — no passwords stored, nothing to breach. Sessions are signed cookies with a "sign out everywhere" revocation list.
- **Secrets are never logged**, never put in error messages, and never sent to the AI layer. Environment variables are validated at startup.

Data-subject rights (GDPR Articles 15, 17, 20)

RIGHT	HOW ORCATRADE HONOURS IT
Access / portability	<code>GET /api/account/export</code> returns a complete JSON copy of everything held — saved plans, events, monitoring alerts and agent memory.
Erasure	<code>POST /api/account/delete</code> hard-deletes plans, alerts and agent memory, pseudonymises event-log entries, and signs the user out — across both data tiers.
Transparency	A public Trust Centre documents data handling, sub-processors, retention and reliability; a cookie-consent banner gates analytics.

The compliance posture

The platform is built UK/EU-first for ICO and GDPR compliance. The Trust Centre is honest about current state — for example, it states plainly that SOC 2 certification is on the roadmap rather than claiming it prematurely (and that honesty is itself enforced by a test). A formal SOC 2 Type II / ISO 27001 path is a funded roadmap item.

Why shareholders should care. Enterprise and channel buyers run a data-protection review before they sign. Having minimised PII, real export/erasure endpoints, an audit trail and a Trust Centre *already in place* shortens that review from a blocker to a checkbox.

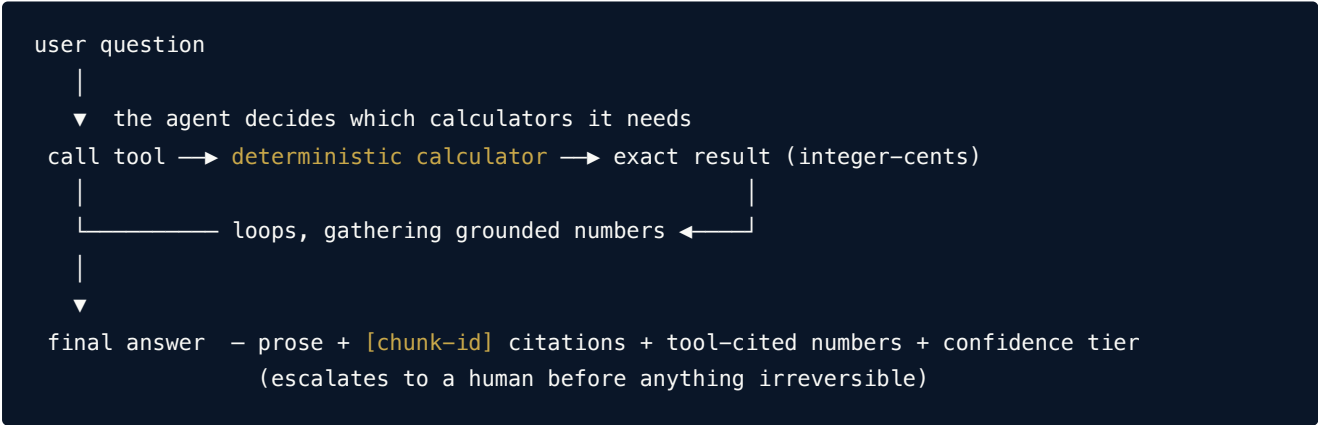
How the agents operate

OrcaTrade fields five domain agents plus a meta-orchestrator and a personal agent. Each is a tool-use loop: the model reasons, calls a calculator as a tool, reads the deterministic result, and continues until it can answer — then writes grounded prose with citations.

The roster

AGENT	OWNS	REPRESENTATIVE TOOLS
Compliance	Duty, AD/CVD, CBAM, EUDR, REACH, CE, sanctions screening	computeLandedCost , checkCbamApplicability , screenCounterparty
Logistics	Mode choice, routing, bonded warehousing, clearance	routing , warehouse
Sourcing	Supplier economics, MOQ/FOB, samples, factory scoring	sourcing , sample
Finance	Payment terms, FX/hedging, working capital, TCO, insurance	fx , tco , finance
Orchestrator	Routes any question across all five domains (merges 14+ tools)	all of the above
Personal	Reasons over the signed-in user's <i>own</i> saved plans, portfolios, deadlines & memory	listMySavedPlans , getMySavedPlanDrift , recallMemory

The tool-use loop



Discipline built into every agent

- **Cite the source.** Every regulatory claim cites a retrieved chunk; every number cites the tool that produced it.
- **Know the limits.** Out-of-scope questions (e.g. medical-device regulation) trigger an honest "not in scope" + human escalation, never a confident guess.
- **Escalate before harm.** High-value or committed actions call a human-review tool rather than proceeding autonomously.
- **Privacy by construction.** The personal agent's tools are locked to the authenticated user — the model cannot request another user's data.

The proactive monitoring agent & agent memory

The agents are not only reactive. The newest capability is a **proactive monitoring agent** — the system watches every customer's saved plans on a schedule and tells them what changed, before they have to ask.

The monitoring agent

A calculator-grounded rules engine re-evaluates each saved plan and portfolio against current data and raises a structured **alert** when something material moves:

SIGNAL	WHAT IT CATCHES
Cost drift	A plan's landed cost has moved $\geq 5\%$ since it was saved — a tariff change, a new anti-dumping measure, or a freight move. The alert names the driver.
FX exposure	An unhedged non-EUR plan where the FX calculator now recommends hedging, with the one-sigma risk figure.
Compliance deadline	A CBAM/EUDR statutory deadline on the user's goods is approaching.
Sanctions update	The consolidated sanctions lists changed — prompting a re-screen of counterparties.

Alerts land in an in-app inbox (with read/dismiss controls) and, where the user opts in, a weekly email digest. Re-scans update existing alerts rather than spamming. Every figure is recomputed by the calculators — the agent never guesses a number.

Agent memory & continuity

The personal agent can now **remember** durable facts a customer states — "our target margin is 35%", "we always clear through Rotterdam", "our main supplier is in Shenzhen" — and recall them in future conversations. Memory is per-user, capped, stored email-hash-only in the durable corpus, and fully covered by the export and erasure endpoints. This turns a series of one-off chats into a relationship that compounds.

Strategic significance. Reactive Q&A is a feature any competitor can copy. A monitoring agent that has watched a customer's plans for a year — and an assistant that remembers how they operate — is a switching cost. This is the beginning of the "import-operations team available 24/7" becoming literally true.

What powers the reasoning

Models

Customer-facing agent reasoning runs on a **frontier Claude model** (Opus-class), chosen because the quality of judgement directly affects whether a customer trusts the answer. Cheaper, faster model tiers are reserved for work that never reaches a customer decision. **Prompt caching** keeps the stable parts of each request (the tool definitions and the system prompt) cached, cutting cost and latency on every call. Every AI call's token usage and cost is recorded to telemetry.

Retrieval — how the agents know the regulations

The regulatory corpus is indexed for **hybrid retrieval**: a keyword search and a semantic vector search are run in parallel and their results fused, so the agent finds the right passage whether the user's wording matches the regulation's wording or not. Each retrieved passage carries a `[chunk-id]` the agent cites. If the embeddings service is unavailable, retrieval falls back to pure keyword search — it degrades, it never fails.

Hybrid

KEYWORD + VECTOR RETRIEVAL, FUSED

Cited

EVERY CLAIM TRACES TO A CHUNK ID

The evaluation moat

Agent quality is not left to chance. A library of eval cases pins the behaviour we require — the right tools called, the right citations, the right escalation, and crucially that *every number in the answer is grounded in a calculator output*. An offline scorer runs free on every code push (a coverage gate that fails the build if the eval suite thins out), and a nightly job runs the full suite against the live model. As models improve, these cases let us upgrade with confidence rather than hope.

The compounding asset. The corpus, the calculators, the eval cases and the customer data are mutually reinforcing: more regulatory coverage improves answers, more eval cases lock in quality, and more customer plans sharpen the calibration of our forecasts against real outcomes.

Trust as a built feature

Tamper-evident audit trail

Every state change is logged with actor, timestamp and before/after state. Beyond a plain log, the audit trail can be exported as a **hash-chained** sequence — each entry cryptographically links to the previous one, so any after-the-fact tampering is detectable. This is the evidence base for due diligence, partner revenue reviews, and regulatory audit.

Live denied-party screening

OrcaTrade screens supplier, buyer and vessel names against the real consolidated sanctions lists — refreshed weekly into Postgres:

19,050 US OFAC SDN	19,761 UK OFSI	1,002 UN SECURITY COUNCIL
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The matching engine is deterministic (name normalisation, token-sort, fuzzy + alias matching) and **safe by design**: it can flag a potential match, but it can never declare a party "clear" — a non-match always carries the advisory that the user must still confirm against the official source. That conservatism is enforced by tests.

Reproducibility & provenance

Every generated plan is stamped with the calculator version, the data as-of date and the generation timestamp — so a result can be reproduced and explained months later. This is what lets us stand behind a number to an auditor.

Other controls

- **Role-based access** with organisations, seats and membership management.
- **Rate limiting** on public endpoints; **token-gated** admin and cron surfaces.
- **Secrets hygiene** — startup validation, never logged, never sent to the AI layer.

Keeping the lights on

Health & status

A structured health endpoint reports the live status of every subsystem — the database, the cache, the tariff cache, email, billing, the AI provider, error tracking, sanctions data and retrieval. A public status page surfaces this, and an uptime check runs on a schedule. The same endpoint underpins paging: a cache outage is a paging condition; a degraded non-critical subsystem is not.

Circuit breakers & structured logging

Every outbound call to a third party — tariff data, email, the AI provider — is wrapped in a circuit breaker with a timeout, retries and a fallback, and traced through a structured logger with request IDs. When an upstream has a bad day, OrcaTrade serves a degraded-but-working experience instead of an error page, and the operator can see exactly what happened.

Error tracking

Production errors flow to a hosted error-tracking service (live today), so regressions surface immediately with full context.

Scheduled jobs (12)

JOB	CADENCE	PURPOSE
monitoring-scan	Weekly	Proactive alerts on saved plans
sanctions-refresh	Weekly	Re-fetch OFAC/OFSI/UN lists
compliance-deadline-reminders	Weekly	CBAM/EUDR deadline emails
regime-change-check	Weekly	Watch EU regulation sources for drift
taric-warm	Daily	Warm the tariff cache for instant answers
calibration-drift-check	Daily	Forecast-vs-actuals drift alerting
weekly-user-digest · plan/portfolio-revision · founder-digest · rag-reindex · db-migrate	Various	Engagement, re-indexing, migrations

Operability is a feature. Because every scheduled job records its last run and outcome, and every subsystem reports health, the team can answer "is everything working?" in seconds — which is what lets a small team run a system enterprise buyers will depend on.

Why the system can be trusted to be correct

Correctness in trade compliance is not optional — a wrong duty rate is a real financial liability. OrcaTrade treats its test suite as the product contract.

2,949AUTOMATED TESTS, ALL
GREEN**2**

NODE LTS VERSIONS IN CI

27

AI EVAL CASES (OFFLINE)

What the tests cover

- **Every calculator** — duty stacking, preferential origin, CBAM/EUDR applicability, FX maths, working capital, TCO — has unit tests pinning the exact numbers.
- **Regression snapshots** guard that a refactor never silently changes a customer-facing figure.
- **The grounding guarantee** is itself tested — the scorer fails any AI answer containing a number that doesn't trace to a calculator.
- **Architectural rules** are enforced in code (e.g. the AI SDK cannot leak into a calculator).
- **Privacy invariants** — that no raw email reaches Postgres, that erasure truly erases — are tested.

Continuous integration

Every push to `main` runs the full suite on two Node versions before it can deploy. The offline AI-eval coverage gate runs in the same pass; a nightly workflow runs the eval suite against the live model. Deployment is automatic once the suite is green.

The discipline compounds. A green 2,949-test suite is not just hygiene — it is what allows a lean team to ship substantial features weekly without breaking the numbers customers rely on. It is also the first thing a technical acquirer or lead investor will inspect, and it holds up.

The content / SEO engine

Alongside the product, OrcaTrade has built a substantial organic-acquisition asset: **658 localised regulatory guides** across English, Polish and German, within 727 indexed pages total — each one a genuine, calculator-backed answer to a real importer's question.

What it is

- One guide per anti-dumping/countervailing measure, per preferential-origin regime, and per compliance regime (CBAM, EUDR, REACH, CE, RoHS, WEEE, batteries, toys, cosmetics, GPSR, PPWR, footwear) — each in three languages.
- A worked-example gallery of real import scenarios (e.g. Chinese e-bikes with 87% combined AD+CVD, Bangladesh apparel under EBA, Turkish steel under A.TR) that each run the real calculators and deep-link into the wizard with inputs pre-filled.
- Full hreflang/sitemap localisation so each language ranks in its own market, with contextual cross-linking between guides.

Why it is a moat

These pages are not thin SEO filler — each is grounded in the same calculators that power the product, so they are accurate, hard to replicate, and convert: a reader who lands on "anti-dumping duty on Chinese e-bikes" is one click from building their own plan. The corpus took sustained effort to build and compounds in search ranking over time. It is a low-cost, durable top-of-funnel that a generic competitor cannot quickly copy.

658

LOCALISED SEO GUIDES

727

INDEXED PAGES TOTAL

3

MARKET LANGUAGES

The path from here — the four pillars

The company's strategy is organised as a re-platforming foundation plus four pillars. Items marked **done** are in production today; **next** items are the funded plan.

Foundation — the re-platform

Done: hybrid pgvector retrieval, Postgres durable corpus via dual-write, frontier-model adoption, prompt caching, observability. **Next:** promote Postgres to primary, durable queues for async work, a full Next.js application shell.

Pillar I — Agent depth & autonomy

Done: frontier-model agents, the proactive monitoring agent (flagship), agent memory, the eval moat. **Next:** document intelligence (ingest & audit a customer's own paperwork), agents that produce the artifact (draft the filing) with human approval, multi-agent delegation.

Pillar II — Platform breadth

Done: sanctions screening (OFAC/OFSI/UN), the compliance calendar, document pre-fill. **Next:** deeper HS10 + rules-of-origin, supplier intelligence & verification, real-time freight & schedules, a trade-finance/insurance marketplace.

Pillar III — Enterprise trust & scale

Done: tamper-evident audit export, reproducibility/provenance, the public Trust Centre, role-based access. **Next:** SCIM provisioning, write-time immutable audit, SOC 2 Type II / ISO 27001, formal SLAs, data-residency options.

Pillar IV — Product surface & go-to-market

Done: the authenticated account surfaces (plans, portfolios, calendar, alerts, screening), the SEO engine. **Next:** the full app shell, onboarding/activation, pricing & packaging for the new value, operational dashboards.

Where the money goes. Investment is directed at the things that either deepen the moat (agent autonomy, regulatory breadth, the data corpus) or remove a buyer objection (SOC 2, SLAs, the app shell). The foundation to support all of it is already in production.

Who is building this

A focused founding team spanning the platform engineering, operations, supply chain, and commercial sides of the business — with the technical depth to build a system this rigorous and the operational and market knowledge to sell it.

ROLE	REMIT
Founder / Platform	Architect of the platform — the calculator engines, the AI agent suite, the compliance overlay, and the test/CI discipline that underpins it all.
Arman Sirin · Chief Operating Officer	Runs operations — turning the platform's recommendations into executed imports; supply-chain delivery, carrier and partner relationships, and the operational backbone that scales client onboarding without scaling headcount.
Chief Supply Chain Officer	Supply-chain execution and the Asia-side sourcing and freight relationships.
Commercial / Client	European importer relationships, onboarding and partnerships.

Operating principle. The team ships continuously — substantial, tested, deployed features on a weekly cadence — while holding a hard line on the correctness and trust guarantees that make the product defensible. This brief is itself a snapshot of a system under active, disciplined development.

How to verify everything in this brief

Every claim here is drawn from the live codebase and the passing test suite as of 25 May 2026 — the architecture, the data model, the agent behaviour, the privacy controls, the sanctions counts, and the test totals are all directly inspectable. Independent technical due diligence is welcomed.

Reference

Technology stack

LAYER	CHOICE
Hosting / deploy	Vercel — auto-deploy on push to <code>main</code> , environment separation
Runtime	Node.js, lean CommonJS codebase; single consolidated serverless router
AI	Anthropic Claude (frontier model for customer-facing reasoning) + prompt caching
Retrieval	Hybrid keyword + vector (pgvector) with reciprocal-rank fusion
Durable data	Neon Postgres (migration-tracked, idempotent schema)
Hot data / cache	Upstash Redis (KV)
Email	Transactional email service (magic links, digests)
Billing	Stripe (subscriptions)
Observability	Hosted error tracking + structured logs + health endpoint + status page
Scheduling	GitHub Actions cron → token-gated dispatcher (12 jobs)
Validation	Schema validation at every external boundary; env validated at startup

The ten calculators

customs-quote	routing-quote	warehouse-quote	sourcing-quote	sample-quote	finance-quote	fx-quote
tco-quote	insurance-quote	returns-quote				

Glossary

TERM	MEANING
HS code	Harmonised System commodity code — determines duty and which regimes apply
AD / CVD	Anti-dumping / countervailing duty — punitive duties that stack on top of base duty
CBAM	EU Carbon Border Adjustment Mechanism — carbon levy + reporting on imports
EUDR	EU Deforestation Regulation — due-diligence on covered commodities
Preferential origin	Trade-agreement rules that reduce or zero duty (EBA, GSP, EVFTA, A.TR, ...)
TARIC	The EU's integrated tariff database
Landed cost	The all-in cost to get goods to the door: goods + freight + duty + VAT + fees
Working capital / CCC	Cash tied up in the trade cycle (inventory + receivables – payables)
RAG	Retrieval-augmented generation — grounding the AI in retrieved regulatory text
Dual-write	Writing a record to both the cache and the durable database

OrcaTrade. The import-operations team, available 24/7 — built on numbers you can audit. · orcatrade.pl